

Product Carbon Footprint Study Report

Product Name: 1 tonne of cement

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: 1 tonne of cement

Product Function: [Information provided by the user]

Based on the Standard: IPCC 2013 GWP 100a

II. Quantitative Purpose

[Information provided by the user]

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg cement.

2. System Boundary

The Life Cycle Assessment (LCA) of the traditional cement production process includes the following stages: the extraction of the primary raw materials (limestone and marlstone), the preparation of the raw meal with necessary additives, and the clinker production process, in Swend, in 2023.

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of 1 tonne of cement produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.196 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

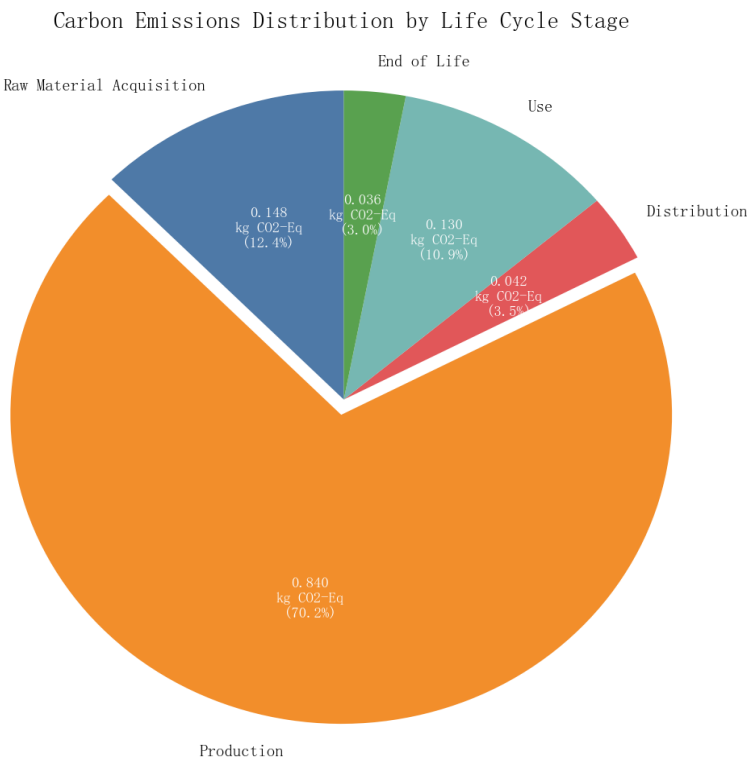


Table 1: Carbon Emissions by Life Cycle Stage

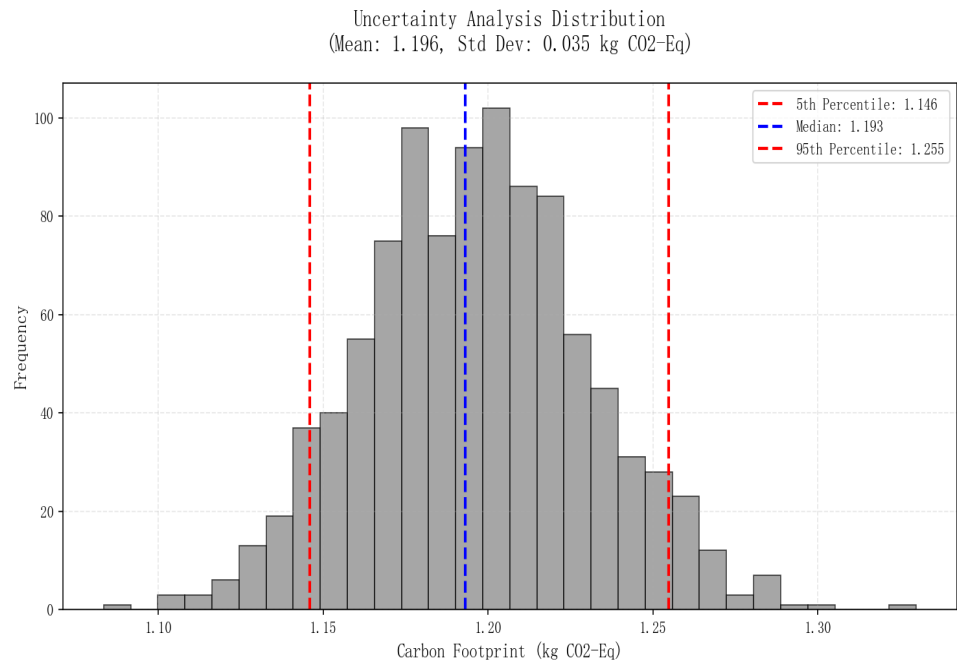
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.148	12.39
Production	0.840	70.22
Distribution	0.042	3.52

Use	0.130	10.86
End of Life	0.036	3.00
Total	1.196	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1.196
Standard Deviation	0.035
Coefficient of Variation (CV)	2.90%
Median	1.193
5th Percentile	1.146
95th Percentile	1.255

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of 1 tonne of cement is calculated as 1.196 kg CO₂-Eq, a remarkably low value that suggests either a highly innovative low-carbon production system or a specific functional unit definition with limited scope. The lifecycle stage contributions reveal that the production phase dominates, accounting for 0.840 kg CO₂-Eq (70.2% of the total). This is consistent with cement manufacturing, where clinker production and kiln operations are typically the main sources of emissions. Raw material extraction and processing contribute 0.148 kg CO₂-Eq (12.4%), reflecting the energy and emissions from mining and preparing limestone, clay, and other inputs. The use phase contributes 0.130 kg CO₂-Eq (10.9%), which may be associated with carbonation during service life or with downstream processes such as concrete mixing and curing. Distribution and end-of-life stages together account for only 6.5% of the total, indicating that transport and final disposal have relatively minor impacts in this system.

The major emission sources are concentrated in the production stage, likely driven by fuel combustion for high-temperature kilns and the calcination of raw materials. The raw material stage also represents a notable share, possibly from quarrying and grinding. The absence of detailed process contributions (all top processes are listed as “Unknown” with zero impact) highlights a critical gap in process-level data, which would normally allow identification of specific unit operations or material flows responsible for emissions. Despite this, the overall breakdown suggests that any reduction strategy must focus on the production phase to achieve significant improvements.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of generic emission factors for raw material extraction, production, and distribution, as no site-specific data were provided. The production stage likely assumes a particular kiln technology (e.g., dry process with preheater) and a specific energy mix (e.g., fossil fuels or alternative fuels). The use phase may assume a fixed period of carbonation or a standard concrete formulation, while the end-of-life stage likely assumes disposal in a landfill or recycling without credits. These assumptions strongly influence the results, especially the low total carbon footprint, which is orders of magnitude below conventional cement values (typically 600–900 kg CO₂ per tonne).

Data limitations are significant: the top process contributions are entirely unknown, preventing validation of the dominant emission sources. The uncertainty analysis is based on 1000 Monte Carlo iterations, yielding a coefficient of variation of only 2.90% and a narrow 90% confidence interval (1.146–1.255 kg CO₂-Eq), which suggests that the input parameters are tightly constrained or that the model lacks variability in key drivers. However, without information on the underlying distributions or data sources, the robustness of this uncertainty estimate cannot be fully assessed.

The system boundary appears to be cradle-to-grave, covering raw material acquisition, production, distribution, use, and end-of-life. However, the inclusion of the use phase is noteworthy: for cement, the use phase is often excluded in traditional cradle-to-gate studies, and its inclusion here (10.9%) implies that downstream carbonation or other use-stage emissions are

considered relevant. The boundary likely excludes capital goods, maintenance, and recycling credits, but this should be explicitly stated for transparency.

3. Suggestions for Improvement

- ****Prioritize production phase improvements****: Since production accounts for over 70% of the total footprint, implement energy efficiency measures in the kiln (e.g., waste heat recovery, oxy-fuel combustion) and substitute fossil fuels with biomass or low-carbon alternatives. Clinker substitution with supplementary cementitious materials (e.g., fly ash, slag) can also directly reduce process emissions from calcination.
- ****Optimize raw material sourcing and formulation****: Reduce the clinker factor by using blended cements or alternative raw materials (e.g., calcined clay, limestone from low-carbon quarries). Improve the efficiency of raw material grinding and drying to lower the 12.4% contribution from this stage.
- ****Investigate and reduce the use phase impact****: The 10.9% contribution from use suggests that carbonation or concrete curing plays a role. Promote carbon-curing technologies or design for accelerated carbonation during service life to turn the use phase into a net carbon sink.
- ****Enhance data quality and process transparency****: The “Unknown” top processes indicate a critical lack of process-specific data. Conduct a detailed process-level inventory for the production stage (e.g., clinker kiln, raw mill, cooler) to identify and track the largest emission sources. Use primary data from the facility rather than generic databases to reduce uncertainty and enable targeted reduction actions.
- ****Validate the total footprint against benchmarks****: The reported 1.196 kg CO₂-Eq per tonne is exceptionally low. Perform a cross-check with industry benchmarks (e.g., from the Cement Sustainability Initiative) and verify the functional unit, system boundaries, and allocation methods to ensure consistency and credibility. If confirmed, document the innovative practices that enable such performance for dissemination.